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Note

An Analytical Study of Mean Roll Radius in Rod Rolling

Youngseog LEE

Plate, Rod & Welding Group POSCO, Technical Research Laboratories, Pohang, P.O. Box 36, Korea. E-mail: pc554162@posco.co.kr

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1. Introduction

In strip (or plate) rolling process, rolling speed at a given pass can be calculated easily by multiplication of roll rpm and roll radius. In rod (or bar) rolling process, however, the rolls have a groove. Consequently, calculating rolling speed becomes difficult because the roll radius is not constant along the roll axis direction. In order to overcome this difficulty, the term "mean roll radius" has been used, which replaces the varying roll radius along the roll groove profile. In this respect, how to determine the mean radius of the grooved roll emerges as a crucial problem in the calculation of rolling speed in rod rolling process.

Wusatowski[1] presented a method for the calculation of the mean roll radius when the maximum spread and the cross sectional area of a deformed workpiece at a pass were known beforehand. Meanwhile, Saito *et al.*[2] proposed a method for the mean roll radius, anchored in the geometry of an incoming workpiece and that of a roll groove profile at a given pass.

Mathematical rationale, however, was not given to the aforementioned methods. While it is acknowledged that the mean roll radius in rod (or bar) rolling is an important parameter in the computation of rolling speed, no systematic study for the determination of mean roll radius has been conducted up to date. This is a basis of this study.

Primary objective of this paper is to develop a mathematics-based equation for computing the mean roll radius so that we can perform a systematic study for the existing models of the mean roll radius.[1,2] Proposed for this purpose was an analytical model for the calculation of the mean roll radius, which can be used in oval-round (or round-oval) pass rolling sequence most commonly employed in rod (or bar) mills. The proposed analytical model is based on the mapping technique that transforms the definition of rolling speed in strip rolling onto that in rod rolling. The analytical model was then applied to a four-pass rolling sequence designed for this study. The mean roll radii calculated by the analytical model were compared with those by the existing models[1,2] and characteristics of these models were discussed.

2. Calculating Models of Mean Roll Radius

In this section, the existing models for the calculation of the mean roll radius are reviewed and the derivation procedure of the proposed analytical model is described.

2.1. The Existing Models

2.1.1. Wusatowski Model[1]

Figure 1 illustrates the physical meaning of the mean roll radius in an oval and a round pass. It is a single value which replaces the varying roll radius along the groove profile. As can be seen, key point is how to determine the deduction value, $\bar{H}$. Wusatowski model can calculate the mean roll radius when one knows information regarding the deformation of workpiece, *i.e.*, the maximum spread and cross sectional area of the deformed workpiece. The mean roll radius is

$$R_{\text{mean}}=R_{\max}-\bar{H}, \quad \text{where } \bar{H}=(A_{\text{p}}/W_{\max}-G)/2. \quad \ldots..(1)$$

A_{p} is the cross sectional area of deformed workpiece at a given pass. Here, the value, $\bar{H}$, is calculated by using the equivalent rectangle approximation method which transforms a curved cross section into a rectilinear one while net cross sectional area is maintained.[3]

2.1.2. Saito Model[2]

Saito *et al.*[2] proposed that the deduction value, $\bar{H}$, could be obtained by knowing the cross points of a roll groove profile and an incoming workpiece at a pass when they are overlapped (**Fig. 2**). Then the mean roll radius is

$$R_{\text{mean}}=R_{\max}-\bar{H}, \quad \text{where } \bar{H}=(A_{\text{h}}/B_{\text{c}}-G)/2. \quad \ldots..(2)$$

A_{h} is the area of the hatched section. Equation (2) is of the same form as Eq. (1) but the value, $\bar{H}$, of Eq. (2) is different from that of Eq. (1). It should be noted that the mean roll radius by Saito model depends on the cross sectional shape of the incoming (undeformed) workpiece. On the other hand, that by Wusatowski model does on that of the outgoing (deformed) workpiece.

2.2. The Analytical Model

The analytical model has been developed based on the mapping which transforms the definition of rolling speed in strip (or plate) rolling onto that in rod (or bar) rolling. Before the analytical model is proposed, reviewed is the mathematical expression of rolling speed being used in a strip rolling. It provides an insight for the determination of the mean roll radius in rod (or bar) rolling. **Figure 3**(a) illustrates the cross sectional shape of workpiece at the roll gap position in strip rolling. The equation for the peripheral speed (=rolling speed), v_{r}, of work rolls contacting the workpiece which has a flat cross sectional shape is expressed as

$$v_{\text{r}}=wR_{\max}, \quad \ldots..(3)$$

where $R_{\max}$ is the maximum roll radius and w is a constant angular velocity of the periphery of the work roll about a rolling axis. It should be noted that the rolling speed, v_{r}, is different from the speed of workpiece, v_{w}. The rolling speed in this study has been defined as the tangential velocity of the work roll rotating at a stand. Thus, in the direction of rolling, the workpiece moves with a speed which is the sum of the rolling speed and the material's flow rate (velocity) due to deformation of the workpiece inside of the roll groove. In other words, $v_{\text{w}}=v_{\text{r}}(1+\phi)$, where ϕ is the forward slip.

Equation (3) is valid for the general case of rigid body

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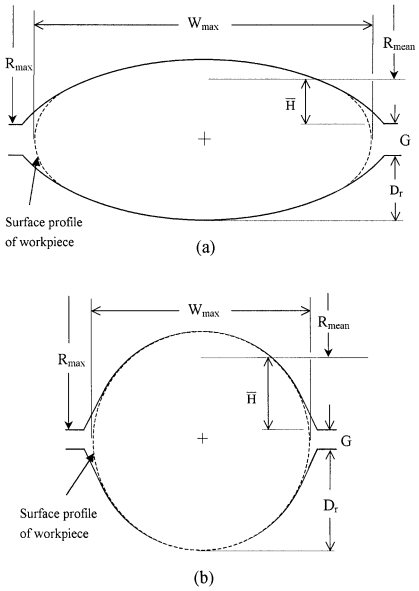


Fig. 1. Geometrical designation for the calculation of the mean roll radius by Wusatowski model[1]: (a) Oval pass, (b) round pass.

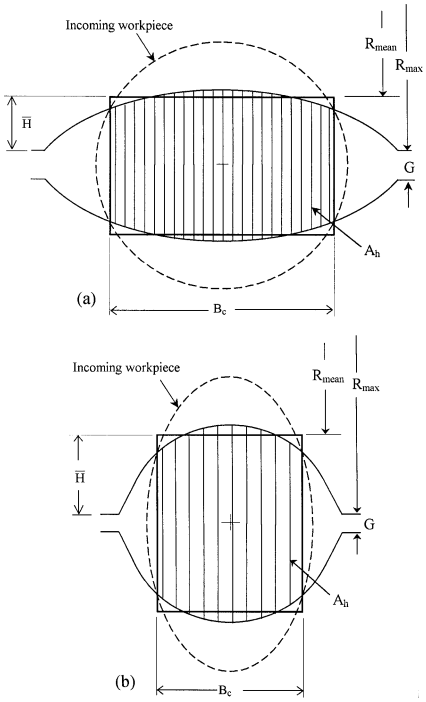


Fig. 2. Geometrical designation for the calculation of the mean roll radius by Saito *et al.*'s model[2]: (a) Round–oval pass, (b) oval–round pass.

Table 1. Mean roll radii calculated for the four-pass rolling sequence (R_{max}=155 mm) when a bar with 60 mm diameter is rolled.

Pass No.	Saito model	Wusatowski model	Analytical model
1st	141.64	142.96	141.26
2nd	136.50	138.76	136.20
3rd	144.99	146.18	144.64
4th	141.46	143.46	141.26

Table 2. Mean roll radii calculated for the four-pass rolling sequence (R_{max}=155 mm) when a bar with 66 mm diameter is rolled.

Pass No.	Saito model	Wusatowski model	Analytical Model
1st	142.35	143.98	142.02
2nd	136.50	138.98	136.32
3rd	144.99	146.22	144.66
4th	141.46	143.47	141.28

rotation about a fixed point. Since $R_1=R_2=\cdots=R_N$ and these are equal to R_{max}, Eq. (3) can be rewritten as

$$v_r=w(R_{max}\cdot N)/N$$
$$=w(R_1+R_2+\cdots+R_N)/N. \quad \text{.....(4)}$$

Equation (4) demonstrates a simple but an important fact. The rolling speed defined in strip rolling process is nothing but the peripheral speed of the work roll contacting the workpiece and can be represented by a sum of finite number of point-wise roll radii along the roll profile at the position of roll gap. Equation (4) becomes a basis in application of the mapping that transforms the definition of rolling speed in strip rolling onto that in rod rolling.

From here onwards, the derivation procedure of the analytical model proposed in this paper is explained in details. Figure 3(b) describes the non-flat cross sectional shape of the deformed workpiece at an oval pass rolling. The mathematical expression for the peripheral speed of the work rolls contacting the workpiece at the roll gap position can be obtained by mapping the definition of rolling speed in strip (plate) rolling onto that in rod (or bar) rolling such that

$$v_r=w(R_1+R_2+\cdots+R_N)/N$$
$$=w\left(\sum_{i=1}^{N}R_i\right)\Big/N.$$
$$=wR_{mean}. \quad \text{.....(5)}$$

Thus, the mean roll radius proposed in this paper is represented by Eq. (6)

$$R_{mean}=\left(\sum_{i=1}^{N}R_i\right)\Big/N. \quad \text{.....(6)}$$

R_1 ($=R_N$) can be calculated once C_x and C_y are known. The point-wise roll radii, $R_2,\cdots,R_{N-1}$, are obtained when the interval, Δl, is calculated beforehand. Referring Fig. 3(b), the expression for the interval, Δl, is chosen as

$$\Delta l=C_x\left(\frac{C_y}{W_f}\right)^{\gamma}, \quad \text{where } \gamma=|W_f/G|. \quad \text{.....(7)}$$

Note that γ is an integer and a non-dimensional constant. It was found that the value of γ does not influence essentially on the calculation result of the mean roll radius if γ is bigger than 4. In a general roll groove design, the ratio of W_f/G is always bigger than 4. The number of the point-wise roll radii along the periphery, N, which is an integer, might be expressed as

$$N=\left|\frac{2C_x}{\Delta l}\right|. \quad \text{.....(8)}$$

Therefore, once C_x and C_y are known, Δl is obtained from Eq. (7) and, in turn, the number of the point-wise roll radii, N, is calculated from Eq. (8). Finally, the mean roll radius, R_{mean}, is computed by Eq. (6). The procedure described above can also be applied to the round pass rolling. Thus, knowing C_x and C_y becomes a vital problem. In order to know C_x and C_y, however, one should be able to predict the surface profile (cross sectional shape) of deformed workpiece at a given pass (**Fig. 4**). The formulation procedure of the surface profile prediction model and its experimental verification are described in the work of Lee and Choi.[4]

3. Results and Discussion

Tables 1 and **2** show the mean roll radii calculated from the existing models (Eqs. (1) and (2)) and the analytical model (Eq. (6)) when a specimen with 60 mm and 66 mm in diameter was rolled through the four-pass rolling sequence

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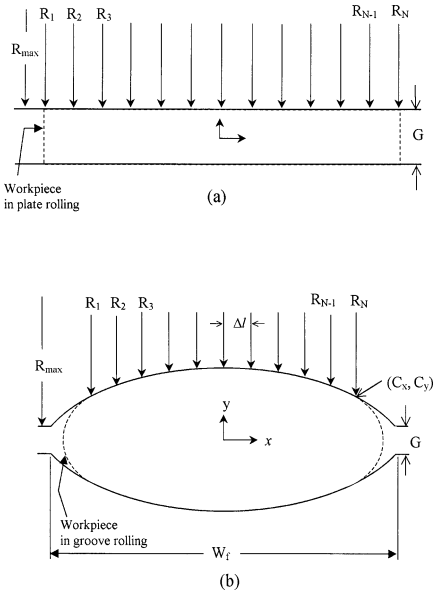


Fig. 3. Schematic of discrete distribution of the roll radii at the position of roll gap (front view): (a) in strip (or plate) rolling, (b) in rod (or bar) rolling.

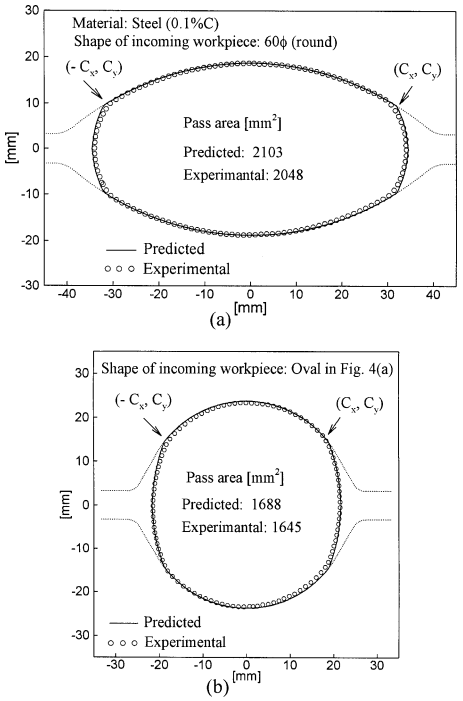


Fig. 4. Predicted and measured surface profiles when a specimen with 60 mm diameter is rolled into (a) 1st pass and (b) 2nd pass described in Fig. 5.

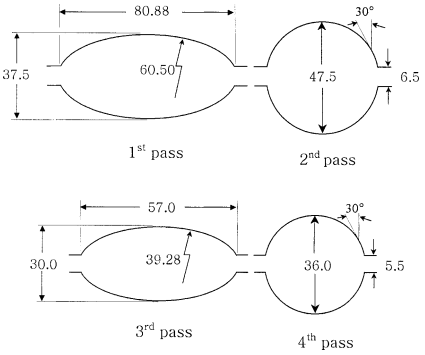


Fig. 5. Four-pass rolling sequences used in hot rod rolling experiment and simulation of the mean roll radius.

described in **Fig. 5**. To understand results correctly, we should focus on the relative differences of the values of mean roll radii, not on its absolute magnitude. The mean roll radii computed by Saito model[2] are very similar with those by the analytical model. On the other hand, the mean roll radii computed by Wusatowski model overestimate slightly those by the analytical model, but the differences with a certain trend are observed as the pass number increases.

In order to investigate the effect of size change of the input specimen on calculation of the mean roll radius, a rolling a bar with 66 mm diameter was simulated. Tables 1 and 2 illustrate the mean roll radii calculated by Saito model is not changed at the passes (2nd, 3rd and 4th). This is due to the definition of Saito model. According to Saito model, the mean roll radius is determined purely from the geometry of the incoming workpiece and the roll groove profile at a pass when they are overlapped. Figure 2(b), for example, shows that the round groove profile and the workpiece with an oval shape rolled at the previous pass are overlapped. For the round pass rolling, we can see that increment of spread to roll axis (roll gap) direction at a previous oval pass does not influence on the calculation of the mean roll radius at a round pass.

If the size of input specimen is increased, the cross sectional area of workpiece during rolling increases. This, in turn, causes the increment of material's flux whose unit is [m^3/sec]. This implies that R_{mean} should be increased to keep up with the increment of material's flux because the rotation of the work roll at a stand is constant. Results shown in Table 2 demonstrate this. The mean roll radii increased as the input specimen size did. Meanwhile, as the pass number increases, the differences of mean roll radii become small. This is because the size of the roll groove profile with the pass number increased is reduced while the maximum roll radius is kept constant.

4. Conclusions

The impetus of this study was to acquire a mathematical expression for the mean roll radius associated with rod (or bar) rolling so that we can perform a systematic study for the existing models[1,2] of mean roll radius currently being used. In order to do this, an analytical model for the calculation of the mean roll radius in round-oval (or oval-round) pass rolling has been developed, which is based on mapping technique. We have compared the mean roll radii calculated from the analytical model with those computed from the existing models. The conclusions are summarized as follows:

(1) Compared with the existing models, the analytical model has a mathematical rationale in formulation since it is based on the mapping that transforms the definition of rolling speed in strip rolling onto that in rod rolling.

(2) Wusatowski model[1] do not have a mathematical ground in formulation but it does not necessary implies that the model was wrong. It can be used for practical usage because of its simplicity in calculation.

(3) Saito model[2] has advantage in the calculation of the mean roll radius due to its easiness, but might have a limitation in a sense; It does not take into account the size change of incoming workpiece, in some cases.

REFERENCES

1) Z. Wusatowski and R. Wusatowski: *Prace Imet*, **4** (1952), 273.
2) Y. Saito, Y. Takahashi, M. Moriga and K. Kato: *J. Jpn. Soc. Technol. Plast*., **24** (1983), 1070.
3) Z. Wusatowski: Fundamentals of Rolling, Pergamon Press, London, (1969) 107.
4) Y. Lee and S. Choi: *ISIJ Int*., **40** (2000), 624.

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